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LLBox API × X360

# What Changed

CHANGES SINCE THE INTEGRATION REFERENCE YOU HOLD



www.llbox.ai

Read alongside the X360 Integration Reference | September 2026

ADDENDUM · VERSION 2.2

# What Changed

Everything that has changed since the guide above. Base URL, authentication and your key are all unchanged, no endpoint has been removed or renamed, and no request field has changed meaning. **Every call you make today keeps working.**

> **The short version.** One change needs code on your side, and it is a new refusal you should read back to the driver. Two are additive and can be adopted whenever suits you. Three are corrections to the guide above, where the described behaviour and the real behaviour had drifted apart.

| # | TYPE | CHANGE | WHAT YOU NEED TO DO |
|---|---|---|---|
| 1 | ACTION | `409` `reason: "ambiguous"` on the reference lookup | Read the message back to the driver |
| 2 | ADDITIVE | `box_photos` in the driver response | Forward them to the driver when ready |
| 3 | ADDITIVE | Optional `expected_delivery_date` at registration, and a new capacity refusal | Adopt when convenient |
| 4 | CORRECTION | `box_status` no longer means what page 7 says | Stop branching on it, if you do |
| 5 | CORRECTION | `otp_expires_at` is now actually enforced | Nothing |
| 6 | CORRECTION | Resetting a box no longer cancels other deliveries | Nothing |

## Why most of this happened

A box used to serve one delivery at a time. That was never true of the physical thing, which holds several parcels, and the mismatch caused real failures: a recipient collecting one parcel cancelled every other driver already on their way, and their codes stopped working at the keypad. A box now serves several deliveries at once, and items 4 through 6 are consequences of putting that right.

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CHANGE 1 · ACTION REQUIRED

# Ambiguous Reference

**GET** `/agent/box/by-reference/:orderReference` Requires API key

Now tolerates how a reference reached the driver, and refuses rather than guessing when that could mean more than one delivery.

Drivers mis-read reference numbers, so the lookup no longer demands a character-perfect match. It ignores punctuation, treats look-alike characters as equal, and accepts a reference clipped at either end.

| TOLERATED | EXAMPLE |
|---|---|
| Punctuation and spacing | `ORD 2026 BIZ → ORD-2026-BIZ` |
| Look-alike characters | `0RD-2026-81Z → ORD-2026-BIZ` |
| Clipped at the front | `4455667 → SHIPMENT-4455667` |
| Extra carrier prefix | `SPL-99-554433 → 554433` |

The folded pairs are `O` / `0`, `I` / `L` / `1`, `S` / `5`, `B` / `8` and `Z` / `2`.

## The new refusal

Where a partial read could match more than one registered delivery, we refuse. Guessing could send a driver to the wrong household with a working code, which is the one outcome worth failing to avoid.

409 The number fits more than one delivery

```
{
  "message": "That number matches more than one expected delivery. Please read the full number again.",
  "reason": "ambiguous"
}
```

**What to do.** Have the agent read the `message` back and ask for the full number. Treating it as a generic failure works, but leaves the driver without the one instruction that resolves it.

An exactly-known reference always wins over an inexact candidate, so this only fires on a genuinely partial or garbled read. Fragments under six characters never match anything.

CHANGE 2 · ADDITIVE

# Site Photos for the Driver

The driver response now carries the photos taken when the box was installed, labelled and ordered so a driver can be walked in from the street to the door. A map pin puts someone on the right road; these get them to the right box.

**GET** `/agent/box/by-reference/:orderReference` — Requires API key

New field `box_photos`. Also present on the fallback lookup `GET /agent/box/:identifier`.

```
"box_photos": [
  { "url": "https://res.cloudinary.com/.../street.jpg",   "category": "street",   "caption": "Second gate on
the left" },
  { "url": "https://res.cloudinary.com/.../door.jpg",     "category": "entrance", "caption": null },
  { "url": "https://res.cloudinary.com/.../box.jpg",      "category": "box",      "caption": null }
]
```

| CATEGORY | WHAT IT SHOWS |
|---|---|
| `street` | The road, so the driver knows they are on the right one |
| `building` | The building from outside |
| `entrance` | The gate or door to approach |
| `box` | The box itself, so it is unmistakable |
| `lock` | The keypad, so the driver knows where the code goes |
| `other` | Anything else worth seeing |

The array is already in the order a driver should see them. At most seven, often fewer, sometimes empty. Where a box has no gallery, it falls back to the single legacy image so the array is never pointlessly empty.

> **These are plain links.** HTTPS, no authentication, no expiry, so they can be forwarded straight into a WhatsApp message and will preview there. The URLs are long and random, which is what keeps them private, so please do not log them anywhere they would be exposed.

`box_photo_url` is unchanged and still present.

CHANGE 3 · ADDITIVE

# Expected Delivery Date

**POST** `/agent/expected-deliveries` Requires API key

New optional request field `expected_delivery_date`. Omit it and nothing changes.

| FIELD | TYPE | NOTES |
|---|---|---|
| `expected_delivery_date` | *string* | Optional. `YYYY-MM-DD`. Must not be in the past, or more than 120 days out. |

It sets how long we hold the reference. Given a date, we hold it until seven days past that day. Without one we assume roughly three days out. That is a real improvement on the old flat fourteen days for everything: a same-day grocery order no longer leaves a live reference for a fortnight, and a shipment weeks away no longer expires before it lands.

## What comes back

```
{
  "expected_delivery_id": 118,
  "status": "pending",
  "box_code": "BOX-FATIMA-01",
  "expected_delivery_date": "2026-11-02",
  "expires_at": "2026-11-09T23:59:59.999Z",
  // present only when that day is getting busy
  "warning": "That makes 4 deliveries expected on 2026-11-02...",
  "deliveries_that_day": 4
}
```

**The warning is advisory.** The registration has already succeeded. It is only worth mentioning to the recipient so they can space their orders out.

## Asking for it

A natural place is straight after the reference: "*roughly when are you expecting it?*" Accept not knowing, and simply omit the field. Same-day platforms make this easy to answer; international shipping rarely does.

CHANGE 3 · ADDITIVE (CONTINUED)

# Box Capacity

A box holds a limited number of parcels on any one day. Capacity is counted per expected day rather than as a running total, because five orders spread across a month were never competing for space.

| DELIVERIES THAT DAY | WHAT HAPPENS |
|---|---|
| Up to 3 | Accepted quietly |
| 4 to 5 | Accepted, with an advisory `warning` in the response |
| 6 or more | Refused with `409` |

`409` That day is full

```
{
  "message": "This box already has 5 deliveries expected on 2026-11-02, which is the most it can take for one day. Pick another day, or collect a parcel first.",
  "code": "BOX_CAPACITY_EXCEEDED",
  "expected_delivery_date": "2026-11-02",
  "deliveries_that_day": 5,
  "soft_limit": 3,
  "hard_limit": 5
}
```

This sits alongside the `REFERENCE_ALREADY_ACTIVE` conflict you already handle. Reading the `message` back is enough; it already names the day and suggests the way out.

> **Worth knowing.** A full Monday does not block Tuesday. If the recipient can name a different day, the same reference will register without complaint.

## Full list of registration refusals

| STATUS | CODE | MEANING |
|---|---|---|
| 404 | — | No account with this phone number |
| 422 | `NO_BOX_LINKED` | Account exists but has no box |
| 422 | `BOX_SELECTION_REQUIRED` | Several boxes, none chosen |
| 403 | `BOX_NOT_OWNED` | That box is not theirs |
| 409 | `REFERENCE_ALREADY_ACTIVE` | Reference live on another account |
| 409 | `BOX_CAPACITY_EXCEEDED` | **New.** That day is full |
| 400 | `INVALID_EXPECTED_DATE` | **New.** Date in the past or too far out |

CHANGES 4–6 · CORRECTIONS

# Corrections to This Guide

### 4 · `box_status` no longer means what page 7 says

Page 7 states that `in_use` means a verification call for a different reference will `409` until it clears, and the example on page 10 shows `"box_status": "in_use"` at claim time. Both are now wrong.

A box serves several deliveries at once. Several drivers can each hold a valid code for the same box simultaneously, which is the normal case when someone has more than one parcel coming. `box_status` now describes only the physical box.

| VALUE | MEANS |
|---|---|
| `available` | Nothing inside. **What you will usually see** on a successful verification. |
| `in_use` | A parcel is inside awaiting collection. Does not block a delivery. |
| `maintenance` | Offline. Claims rejected with `403`. Unchanged. |
| `deactivated` | Retired. Claims rejected with `403`. Unchanged. |

> **Please do not branch on `box_status`.** It is informational. The `409` for a genuinely unavailable box still exists but is now rare, and carries `reason: "at_capacity"`.

### 5 · `otp_expires_at` is now true

Delivery codes previously had no time limit at the box. The field was returned, but the keypad ignored it, so any code ever issued stayed able to open its box indefinitely. Codes now last three days and the keypad enforces it.

`POST /agent/otp/validate` correspondingly returns `{"valid": false, "reason": "expired"}` when a code has run out, where previously it could not.

> **A driver does not lose the delivery.** If a code expires unused, the delivery returns to waiting and a later driver can claim it again with a fresh code. Carriers retry, and a failed first attempt no longer burns the reference.

### 6 · Resetting a box no longer cancels other deliveries

`POST /agent/box/:box_id/reset` previously expired every in-progress delivery on the box and revoked every unused code. With one delivery at a time that was survivable. With several it meant a recipient collecting one parcel silently cancelled every other driver already on the way, and their code stopped working at the keypad.

Reset now frees the box and nothing else. Deliveries still heading there keep their status and their codes.

REFERENCE · UPDATED

# Driver Response in Full

The complete success body as it stands today, replacing the example on page 10. New and changed fields are called out below it.

**GET** `/agent/box/by-reference/:orderReference` — Requires API key

200 Resolved

```
{
  "box_id": 7,
  "box_code": "QA-MEMBERSHIP-E2E",
  "box_status": "available",          // see addendum p.6
  "box_photo_url": "https://res.cloudinary.com/.../box.jpg",
  "box_photos": [                     // new, see addendum p.3
    { "url": "https://...", "category": "street", "caption": "Second gate on the left" }
  ],
  "address": "Al Olaya, Riyadh",
  "address_arabic": null,
  "latitude": 24.7136,
  "longitude": 46.6753,
  "google_maps_link": "https://maps.google.com/?q=24.7136,46.6753",
  "recipient_name": "Khalid Baig",
  "recipient_whatsapp_number": "+966500000000",
  "delivery_session_id": "42",
  "otp": "738921",
  "otp_expires_at": "2026-11-05T13:45:00.000Z",  // now enforced, see addendum p.6
  "pending_delivery": null,
  "instructions": "Enter this code on the box keypad to open it."
}
```

| FIELD | STATUS |
|---|---|
| box_photos | NEW Labelled site photos, in the order to show them |
| box_status | CHANGED Describes the physical box only. Do not branch on it. |
| otp_expires_at | CHANGED Now genuinely enforced at the keypad |
| box_photo_url | Unchanged, still present |

Every other field is byte-for-byte as documented on page 10.

REFERENCE · UPDATED

# Refusals and Checklist

## Every way the driver lookup can refuse

| STATUS | REASON | MEANING |
|---|---|---|
| 404 | `not_found` | No such reference registered |
| 404 | `expired` | Its hold window passed |
| 404 | `already_matched` | A driver holds a code for it |
| 404 | `already_completed` | Already delivered |
| 404 | `cancelled` | The recipient cancelled it |
| 409 | `ambiguous` | **New.** Fits more than one delivery. Ask for the full number. |
| 409 | `at_capacity` | **New.** Box genuinely cannot take another. Rare. |
| 403 | `maintenance` | Box offline. Unchanged. |
| 403 | `deactivated` | Box retired. Unchanged. |

## Integration checklist

| | ITEM |
|---|---|
| ☐ | Agent handles `409 ambiguous` by reading the message and asking for the full number |
| ☐ | Nothing branches on `box_status` |
| ☐ | An unexpected `reason` value degrades gracefully rather than erroring |
| ☐ | *Optional.* `box_photos` forwarded to the driver over WhatsApp |
| ☐ | *Optional.* Recipient asked for an expected day at registration |
| ☐ | `409 BOX_CAPACITY_EXCEEDED` message read back to the recipient |

**Testing.** There is still one environment, production, as page 5 explains. Use a clearly fake, greppable reference such as `X360-TEST-<date>`. For an end-to-end rehearsal, LLBox can now simulate a physical box from its admin panel, so a full journey can be walked without hardware on site. Ask and we will set up a window.

**Questions.** Anything unclear in this addendum, tell us which page and we will fix the document rather than explain it once over a call.

ADDENDUM 2 · WHATSAPP MANAGEMENT

# Managing the Box by Text

Today the bot can register a delivery. It has no way to know whether the person texting has an LLBox at all, and no way to help them with anything else. These endpoints close that gap.

> **Entirely additive.** Nothing in the guide or in Addendum 1 changes. Your current integration keeps working untouched.

## Start every message with one call

`POST /agent/whatsapp/identify` answers three questions at once: is this number ours, who is it, and what is worth offering them right now. Call it on any inbound message before deciding what to say.

## What we trust, and what we do not

**Trusted:** that you correctly report which number sent the message. WhatsApp verifies phone ownership at the platform level, and you authenticate with your API key, so a caller cannot claim to be someone else without both compromising that key and lying about the sender.

**Not trusted:** that a known number may touch any given box. A box is shared by up to four people, so every action separately confirms the account is linked to the box or delivery it names. Identity is not authorization.

> Two consequences worth knowing. An unknown number gets a flat `404` with no detail, so these cannot be used to test whether someone has an LLBox. And asking about a delivery on another household's box returns `404` rather than `403`, so a wrong id never confirms that their delivery exists.

All of these are `POST`, including the read-only ones, so a phone number travels in a request body rather than a URL that would end up in access logs and proxy traces.

| ENDPOINT | PURPOSE |
|---|---|
| `whatsapp/identify` | Is this our customer, and what to offer |
| `whatsapp/deliveries` | What is on the way |
| `whatsapp/deliveries/cancel` | Take back a mistaken registration |
| `whatsapp/code` | Issue or replace a box code |
| `whatsapp/box-status` | Has it arrived, how busy is the box |
| `whatsapp/notify-recipient` | Reach the customer when a driver is stuck |

ADDENDUM 2 · ENDPOINT

# Identify the Sender

**POST** `/agent/whatsapp/identify` — Requires API key

The first call on any inbound message.

```
{ "whatsapp_phone": "+966500000001" }
```

**200** Recognised

```
{
  "user_found": true,
  "recipient_name": "Fatima Al-Otaibi",
  "boxes": [
    { "box_id": 5, "box_code": "BOX-FATIMA-01", "nickname": "Al Malqa, Riyadh",
      "status": "available", "your_role": "owner" }
  ],
  "available_actions": ["register_delivery", "list_deliveries", "cancel_delivery", "get_code",
"box_status"],
  "active_deliveries": 2,
  "parcel_awaiting_collection": false,
  "summary": "You have 1 delivery registered, 1 with a driver on the way."
}
```

`available_actions` is the useful part: offer only what applies. `cancel_delivery` appears only when something is actually cancellable, so the bot never offers to cancel to someone with nothing registered. `summary` is written to be read aloud or sent as-is.

`your_role` is `owner` for the person the box belongs to and `member` for someone they added. Both can do everything in this addendum.

**404** Not one of ours

```
{ "message": "No LLBox account is registered to this number", "user_found": false }
```

> Treat this as "not a customer" rather than an error. Nothing else is returned, deliberately.

ADDENDUM 2 · ENDPOINTS

# Deliveries

**POST** `/agent/whatsapp/deliveries` — Requires API key

What is heading to their boxes. Add `"include_finished": true` for completed and cancelled ones too.

```
{
  "deliveries": [
    {
      "delivery_id": 118,
      "reference": "SHEIN-88213004",
      "status": "pending",
      "box_code": "BOX-FATIMA-01",
      "expected_delivery_date": "2026-11-02",
      "registered_by_them": true,
      "can_cancel": true
    }
  ]
}
```

Scoped to the boxes they belong to rather than what they personally registered, the same rule the mobile app uses, so everyone sharing a box sees the same picture. `registered_by_them` distinguishes their own from a housemate's, and `can_cancel` saves you having to know the rule.

**POST** `/agent/whatsapp/deliveries/cancel` — Requires API key

Only while pending. Once a driver holds a code they are on their way.

```
{ "whatsapp_phone": "+966500000001", "delivery_id": 118 }
```

| STATUS | CODE | MEANING |
|---|---|---|
| 200 | – | Cancelled |
| 200 | – | Already cancelled, said plainly rather than as an error |
| 409 | DELIVERY_NOT_CANCELLABLE | A driver has a code, or it is finished |
| 404 | DELIVERY_NOT_FOUND | Not on a box this number is linked to |

ADDENDUM 2 · ENDPOINTS

# Codes and Box Status

**POST** `/agent/whatsapp/code` — Requires API key

Issues a box code, the same thing the app's New Code button does.

```
{ "whatsapp_phone": "+966500000001", "box_id": 5, "delivery_id": 118 }
```

**With** `delivery_id` it becomes that delivery's code, which covers a driver who rang the recipient directly instead of you, or one whose own code will not work. It replaces and revokes the driver's previous code, and entering it at the box marks that delivery delivered.

**Without** `delivery_id` it is a plain guest code for letting someone in. It opens the box and closes no delivery.

`box_id` may be omitted when the account has exactly one box. With more than one you get `422 BOX_SELECTION_REQUIRED` and the list to ask from, the same shape as registration.

```
{
  "message": "Code created and linked to the delivery",
  "code": "738921",
  "expires_at": "2026-11-05T13:45:00.000Z",
  "linked_delivery": { "delivery_id": 118, "reference": "SHEIN-88213004", "status": "matched" }
}
```

**The code is returned once and cannot be retrieved afterwards.** If the recipient loses it, issue another. Please do not persist it anywhere it would outlive the conversation.

**POST** `/agent/whatsapp/box-status` — Requires API key

Answers "is my parcel there yet?" without the recipient opening the app.

```
{
  "boxes": [
    { "box_code": "BOX-FATIMA-01", "parcel_inside": true,
      "deliveries_expected_today": 2, "active_deliveries": 3,
      "comfortable_per_day": 3, "most_per_day": 5 }
  ]
}
```

ADDENDUM 2 · ENDPOINT

# Reaching the Customer

**POST** `/agent/whatsapp/notify-recipient` Requires API key

Notifies the customer in their app when a driver is stuck.

**Why this exists.** `recipient_whatsapp_number` comes back only on a **successful** driver lookup. In exactly the situations where a driver most needs the customer — the reference expired, the parcel will not fit, the code will not open the box — you may have no way to reach anyone at all.

```
{
  "order_reference": "CRAZY1232324",
  "issue": "parcel_too_large",
  "note": "Box is full, I left it by the gate"
}
```

| ISSUE | USE WHEN |
|---|---|
| `reference_expired` | The hold window passed; it needs registering again |
| `code_not_working` | The keypad is refusing the code |
| `parcel_too_large` | It will not fit, or the box is already full |
| `cannot_find_box` | At the address and cannot see the box |
| `box_full` | Nothing fits until something is collected |
| `other` | Anything else — put the detail in `note` |

`note` reaches the customer in the driver's own words, so please use it. "At the blue gate, no box visible" is worth far more than a category alone.

200 Someone was told

```
{
  "notified": true,
  "people_notified": 2,
  "message": "The customer has been notified in their LLBox app."
}
```

200 Nobody to tell

```
{ "notified": false, "message": "Nothing to notify anyone about for that reference." }
```

**Four things worth knowing.** Everyone sharing that box is notified, since a housemate may be the one at home. No phone number is ever returned: LLBox does the notifying, not you. An unknown reference returns `notified: false` and nothing else, so this cannot be used to test which references exist. And unlike the rest of this section it takes no `whatsapp_phone` — the caller is a **driver**, whose number we neither know nor need.

ADDENDUM 2 · GUIDANCE

# The App and the Bot Stay in Step

Everything in this addendum writes through the same data the mobile app reads. A delivery registered by text appears in the app immediately, one cancelled in the app disappears from the bot's listing, and a code issued either way works at the same keypad. There is no separate WhatsApp state that can drift.

That is covered by tests rather than assumed. The suite registers by text and asserts the app sees it, cancels in the app and asserts the bot no longer lists it, and drives a code issued by text through the physical keypad path.

## A shape for the conversation

| STEP | THE PERSON SAYS | CALL |
|---|---|---|
| 1 | Anything at all | `whatsapp/identify` |
| 2 | Number not recognised | Offer to help them sign up. Reveal nothing else. |
| 3 | Recognised | Greet by name, read `summary`, offer only `available_actions` |
| 4 | "Register a delivery" | Ask the reference, then optionally the day, then `agent/expected-deliveries` |
| 5 | "Where is my parcel" | `whatsapp/deliveries` |
| 6 | "Has it arrived" | `whatsapp/box-status` |
| 7 | "Cancel one" | `whatsapp/deliveries` to list, then `.../cancel` |
| 8 | "The driver's code isn't working" | `whatsapp/code` with the `delivery_id` |
| 9 | A driver is stuck and needs the customer | `whatsapp/notify-recipient` |

## Worth saying out loud

- Offer only what is in `available_actions`. It already accounts for whether they have a box, and whether anything is cancellable.
- Read our `message` back on a refusal. They are written for a person, and usually name the way out.
- An unexpected field or status should degrade gracefully. This document will grow.

**Questions.** Anything unclear here, tell us which page and we will fix the document rather than explain it once over a call.